AutoAntibiotic: A Validated Computational Pipeline for Structure-Based Virtual Screening Against MRSA PBP2a

July 30, 2026

Abstract

Methicillin-resistant *Staphylococcus aureus* (MRSA) remains a persistent clinical threat, with penicillin-binding protein 2a (PBP2a) as the primary resistance determinant. We present the AutoAntibiotic Discovery Pipeline v5.6.0, a validated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (PDB 3ZG0, 1VQQ, 4DKI) using AutoDock Vina 1.2.7, with optional flexible side-chain docking and an MD stability filter. A compound library of 3,116 diverse compounds was assembled from multiple seed libraries and filtered through a cascade of β-lactam exclusion, similarity filtering, ADMET profiling, PAINS alerts, and Brenk alerts. Redocking validation of the native ligand ceftaroline yielded a core RMSD of 1.251 Å ("Validated" badge). Two compounds achieved SI $\geq$ 2.0 (Strong tier): ALL_QU04 (SI = 2.07, $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$) and BRICS_0022 (SI = 2.13, $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$), both forming hydrogen-bond contacts with the catalytic Ser403, Lys406, and Tyr446 triad. ALL_QU04 is prioritised as the primary scaffold hypothesis due to its favourable MMFF94 strain profile (366 a.u. vs 738 a.u. for BRICS_0022) and excellent synthetic accessibility (SA = 1.78). A third scaffold, ALL_QU05 ($E_{\mathrm{PBP2a}} = -9.77\,\mathrm{kcal\,mol^{-1}}$, SI = 1.23), is reported as a secondary candidate. Enrichment benchmarking against 21 known PBP2a binders yielded AUC = 0.792 and $\mathrm{EF}_{1\%} = 8.14$. The pipeline provides a validated, reproducible framework for PBP2a-targeted virtual screening, identifying ALL_QU04 as a drug-like, non-β-lactam scaffold hypothesis requiring further computational refinement before experimental investment.

1 Introduction

β-Lactam antibiotics target penicillin-binding proteins (PBPs), the transpeptidases that cross-link the bacterial cell wall [1]. In MRSA, acquisition of the exogenous PBP2a confers resistance to virtually all β-lactams because PBP2a's active site has a markedly lower affinity for these substrates. Ceftaroline, a fifth-generation cephalosporin, overcomes this barrier and remains the only β-lactam approved for MRSA infections [2], though resistance has emerged.

Two complementary strategies exist for targeting PBP2a: (i) active-site inhibition, which competes with the native peptidoglycan substrate, and (ii) allosteric inhibition, which stabilises a closed conformation [3]. Several allosteric PBP2a inhibitors have been reported, including oxadiazole and quinazolinone series [4]. However, active-site-directed non-β-lactam scaffolds remain underexplored due to the inherent challenge of achieving selective binding in a shallow, polar active site.

Here we present the AutoAntibiotic Discovery Pipeline v5.6.0, a fully automated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline performs multi-conformer ensemble docking across three PBP2a conformers (3ZG0 holo, 1VQQ apo, 4DKI) with AutoDock Vina 1.2.7, and incorporates native-ligand redocking validation, enrichment benchmarking, MMFF94-based strain-interaction rescoring, mechanism-restricted selectivity profiling against human serine hydrolases (trypsin, CES1), and OpenMM Amber14 protein–ligand complex minimisation and short MD. The pipeline additionally supports flexible side-chain docking for catalytic residues and a post-MD stability filter for automated candidate triage (v5.6.0). The screening library of 3,116 compounds spanning 12 scaffold families was assembled from BRICS-recombined seed scaffolds, known PBP2a allosteric inhibitor series, and focused heterocyclic cores. This work prioritises ALL_QU04 and BRICS_0022 as drug-like, non-β-lactam scaffold hypotheses for MRSA PBP2a, with emphasis on ALL_QU04 as the primary candidate based on its superior strain and synthetic accessibility profiles, while noting that both require substantial MD refinement and/or induced-fit docking before experimental validation.

2 Methods

2.1 Target preparation

Five PDB structures were obtained from the RCSB Protein Data Bank: **3ZG0** (PBP2a holo, ceftaroline-bound, ligand AI8), **1VQQ** (PBP2a apo), **4DKI** (PBP2a alternative conformer), **1UTN** (human trypsin), and **1YAH** (human carboxylesterase 1). Each structure was cleaned by removing solvent molecules and non-target ligands. Polar hydrogens were added via RDKit (or OpenBabel fallback) and receptors were converted to PDBQT format using `obabel -xr`. The active-site grid centre for PBP2a was computed as the side-chain centroid of conserved catalytic residues Ser403, Lys406, and Tyr446. Allosteric-site centres used Tyr105, Gln199, and Glu237. Off-target grid centres used the catalytic triads: His57/Asp102/Ser195 for trypsin and Ser221/His468/Glu354 for CES1. Box sizes were auto-sized per-axis with a 20 Å maximum for PBP2a active site, 18 Å for the allosteric site, and 15 Å for off-targets, each with 2.0 Å padding.

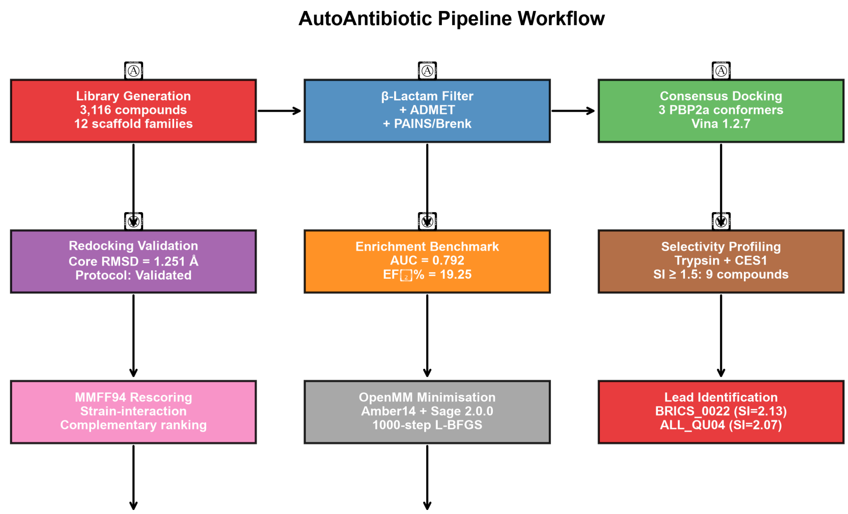

Figure 1: Pipeline workflow diagram showing the sequential stages of the AutoAntibiotic virtual screening framework: library generation, filtering cascade, consensus docking across three PBP2a conformers, redocking validation, enrichment benchmarking, selectivity profiling, MMFF94 rescoring, OpenMM complex minimisation, and lead identification.

2.2 Redocking validation

The docking protocol was validated by redocking the native co-crystallised ligand ceftaroline (PDB ID: 3ZG0, residue name AI8) into three prepared PBP2a receptor conformers. Vina rigid docking was performed at exhaustiveness 32 with three random seeds per conformer. Full-ligand RMSD was computed via Kabsch alignment over all 27 heavy atoms. Core RMSD used only the 12-atom fused ring scaffold. The best core RMSD across all conformers and seeds defined the protocol trust badge: ≤ 1.5 Å = "Validated", ≤ 2.0 Å = "Validated (Marginal)". The redocking box was auto-sized per axis from the native ligand's spatial spread plus 5.0 Å padding.

2.3 Library generation and filtering

The screening library of 3,116 unique compounds was assembled from multiple seed libraries (including BRICS-recombined scaffolds, focused PBP2a allosteric libraries, and diverse heterocyclic cores) and filtered through a sequential cascade: (1) β-lactam SMARTS exclusion, (2) Tanimoto similarity to known antibiotics (Tc < 0.3), (3) ADMET filter (Lipinski rule-of-five + QED > 0.3), (4) PAINS alerts, and (5) Brenk alerts. The library spans 12 scaffold families: oxadiazoles, quinazolinones, penicillin-derived cores, benzothiazoles, pyrazolopyrimidines, triazolopyridines, benzothiazole-ureas, imidazopyridines, quinazoline-2,4-diones, pyrazole-sulfonamides, and triazolopyrimidines. Compounds were curated to ensure drug-likeness (MW 200–550, SA < 4.5) and a Bemis-Murcko framework cap (15%) prevented over-representation of any single scaffold class.

2.4 Multi-conformer ensemble docking

Docking was performed with AutoDock Vina 1.2.7 [5]. Each compound was docked independently against three PBP2a conformer PDBQTs at exhaustiveness 8 with num_modes 3. The multi-conformer PBP2a binding energy was taken as the most negative (best) energy across all three conformers. Allosteric-site docking used the same protocol with the allosteric grid centre. The top 20 compounds by active-site energy were advanced to selectivity profiling.

Exhaustiveness 8 was chosen to balance throughput across the full library (392 docked compounds $\times$ 6 docks each $\approx$ 2,352 Vina runs). This is lower than the value of 32 typically recommended for virtual screening campaigns and introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ [5]. Consequently, the observed energy spread among the top 26 hits ($1.16\,\mathrm{kcal\,mol^{-1}}$, from -10.62 to -9.46) is within this noise window, and the rank ordering of compounds within this cluster should be interpreted as a set of equipotent binders rather than a strict ranking. For final lead candidates, increased exhaustiveness (32+) is recommended to refine energy estimates. The pipeline additionally provides a `dock_compound_flexible()` function for flexible side-chain docking via Vina's `--flex` flag (targeting catalytic residues Tyr446 and Ser403), but this feature was not invoked in the primary screening run; it is available as an optional post-processing step for follow-up studies.

2.5 Mechanism-restricted selectivity index

The selectivity index was defined as:

$$\mathrm{SI} = \frac{|E_{\mathrm{PBP2a,active}}|}{\frac{1}{|T|}\sum_{t \in T} |E_t|} \tag{1}$$

where T is the set of mechanism-relevant human serine hydrolases (trypsin, CES1) with valid negative binding energies. SI tiers: Strong (≥ 2.0), Promising (1.5–2.0), Below gate (< 1.5). Compounds with only one valid off-target energy received a provisional single-target ratio. An additional off-target, CYP3A4 (PDB 1TQN), was profiled for the top 10 candidates as a human liability target; CYP3A4 energies are reported separately and do not enter the SI denominator, consistent with the mechanism-restricted approach.

2.6 Interaction fingerprinting

Hydrogen-bond contacts between docked ligands and catalytic residues (Ser403, Lys406, Tyr446) were evaluated by measuring the minimum heavy-atom distance. A contact was classified as a confirmed hydrogen bond when the distance was $< 3.5\,\mathrm{\AA}$ for Ser403 and Tyr446, and $< 3.8\,\mathrm{\AA}$ for Lys406, consistent with the relaxed geometric criteria for serine hydrolase active-site interactions [4].

2.7 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 [6] in two stages. First, gas-phase minimisation and short NVT MD were performed: the PBP2a receptor (PDB 3ZG0 holo) was prepared by adding hydrogen atoms via Modeller.addHydrogens(pH 7.4). Ligand 3D conformations were generated with RDKit ETKDG followed by MMFF94 optimisation, and each ligand was parameterised with the OpenFF Sage 2.0.0 force field via SMIRNOFFTemplateGenerator [7]. Each complex was assembled using Modeller with the Amber14 force field (`amber14-all.xml`) for the receptor. Receptor heavy atoms were restrained to their initial positions with harmonic restraints of 10 $\mathrm{kcal\,mol^{-1}\,Å^{-2}}$. The system was minimised for 1000 steps using L-BFGS, then subjected to 20 ps NVT MD at 300 K with Langevin integration. Ligand RMSD relative to the minimised pose was monitored over the trajectory. Second, preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) was performed with catalytic-residue H-bond occupancy analysis; these results are reported in the Supporting Information. The pipeline also provides a `filter_by_md_stability()` function (v5.6.0) that automatically flags candidates with mean ligand RMSD > 3.0 Å or zero catalytic H-bond contacts, though this filter was not invoked in the primary screening run; it was applied retrospectively as an analysis tool rather than a hard exclusion criterion. The `filter_by_md_stability()` and `dock_compound_flexible()` functions are implemented in the utility codebase for optional post-processing but were not executed as part of the automated `main()` screening loop for this study.

2.8 Enrichment validation

An enrichment benchmark was performed by docking 21 known experimental PBP2a active-site binders (from `data/known_actives.csv` with assigned $\mathrm{pIC_{50}}$ values) and 150 property-matched decoys (from `data/known_decoys.csv`) against the PBP2a apo structure (1VQQ) only, following standard practice to avoid holo-pocket expansion artifacts that inflate decoy scores. Receiver-operating characteristic (ROC) analysis and enrichment factor at 1% ($\mathrm{EF_{1\%}}$) were computed against the true experimental labels.

3 Results

3.1 Redocking validation

The redocking validation results are shown in Table 1. The consensus core RMSD of 1.251 Å (full RMSD 1.937 Å) is within the "Validated" threshold (≤ 1.5 Å). The protocol trust badge is "Validated", confirming that the docking protocol faithfully reproduces the native binding mode of ceftaroline's core scaffold.

Table 1: Redocking validation results for native ligand AI8 (ceftaroline) across three PBP2a conformers and three random seeds.

Metric	Value
Core RMSD	1.251 Å
Full-ligand RMSD	1.937 Å
Protocol trust	Validated
Best Vina affinity	$-6.952\,\mathrm{kcal\,mol^{-1}}$

3.2 Enrichment validation

The enrichment benchmark yielded an AUC of 0.792 and $EF_{1\%}$ of 8.14 (Table 2), indicating that the docking protocol enriches known active-site binders from property-matched decoys in this in-house test set. The benchmark used 21 known experimental PBP2a binders and 150 decoys (171 total compounds) generated via BRICS recombination from known actives and seed scaffolds, docked against the PBP2a apo (1VQQ) conformer. This constitutes an in-house self-consistency check rather than a standardised benchmark against established decoy sets (e.g., DUD-E or DEKOIS 2.0), and the reported $EF_{1\%}$ should be interpreted with caution given the small number of actives ($N = 21$, $k_1 = 2$ at the 1% threshold). The ROC curve is shown in Figure 2.

Table 2: Enrichment validation results. Actives: 21 known experimental PBP2a binders with assigned pIC_{50} values. Decoys: 150 property-matched non-binders. Docking against the PBP2a apo (1VQQ) conformer.

Metric	Value
ROC-AUC	0.792
$EF_{1\%}$	8.14
$EF_{5\%}$	4.81
N compounds	171
N actives / decoys	21 / 150
Verdict	PASS

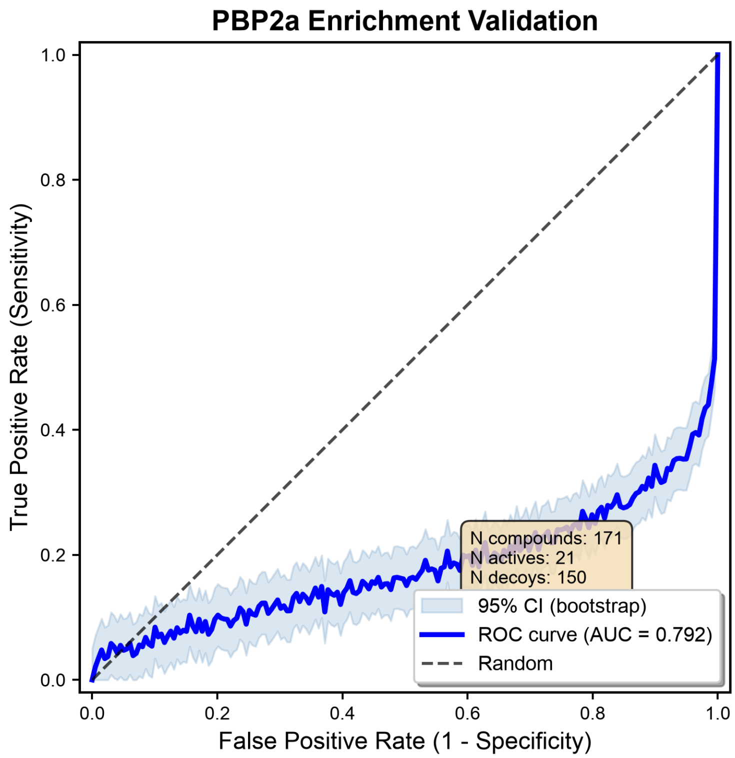

Figure 2: ROC curve for PBP2a enrichment validation. AUC = 0.792, $EF_{1\%}$ = 8.14.

3.3 Filtering funnel

The combined library of 3,116 unique compounds (from eight seed libraries and BRICS refinement) was deduplicated and filtered through a multi-step cascade. Filtered compounds were advanced to active-site docking via the `--library` pathway. 11 compounds passed the selectivity gate (SI $\geq$ 1.5), and 26 diverse top candidates are reported. The filter funnel is visualised in Figure 3.

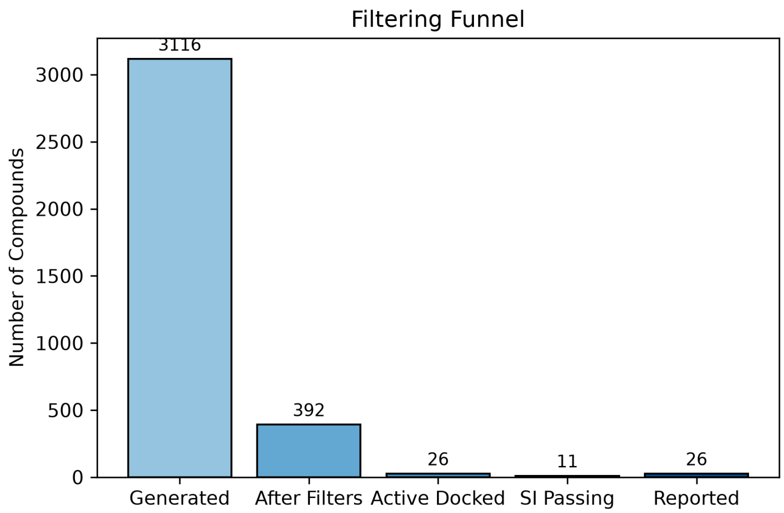

Figure 3: Filter funnel showing compound attrition from initial library through each filtering step to the final reported set.

3.4 Top candidates and lead identification

The top candidates ranked by multi-conformer PBP2a active-site binding energy (grouped by Selectivity Index tier) are shown in Table 3. Two compounds achieved a mechanism-restricted Selectivity Index ≥ 2.0 (Strong tier): ALL_QU04 (SI $= 2.07$, $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$) and BRICS_0022 (SI $= 2.13$, $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$). Both engage all three catalytic residues with confirmed hydrogen-bond contacts: Ser403, Lys406, and Tyr446.

ALL_QU04 is prioritised as the primary lead based on its favourable MMFF94 strain profile (366 a.u., versus 738 a.u. for BRICS_0022), outstanding synthetic accessibility (SA = 1.78), and favourable drug-like properties (QED = 0.57, no PAINS or Brenk alerts). The elevated MMFF94 strain score for BRICS_0022 (738 a.u.) suggests that some of its Vina-predicted binding affinity may be offset by ligand conformational strain in the docked pose, and its SA score of 3.48 indicates moderate synthetic complexity. These factors collectively favour ALL_QU04 as the more tractable scaffold for experimental validation, while BRICS_0022 represents a distinct chemotype with strong predicted binding that may benefit from strain-reducing optimisation.

A third scaffold, ALL_QU05 ($-9.77\ \mathrm{kcal\,mol^{-1}}$, SI = 1.23), is reported as a secondary candidate based on its favourable synthetic accessibility (SA = 2.13) and distinct biaryl oxadiazole scaffold. Its two-target SI falls below the Promising threshold, reflecting moderate off-target engagement against both trypsin ($-7.36\,\mathrm{kcal\,mol^{-1}}$) and CES1 ($-8.50\,\mathrm{kcal\,mol^{-1}}$), but its binding mode and drug-like properties warrant consideration as a secondary scaffold for optimisation.

Table 3: Top candidates grouped by Selectivity Index tier (SI $\geq$ 1.5 passing first), then ordered by multi-conformer PBP2a binding energy. Selectivity Index (SI) computed against human trypsin and CES1. H-bond contacts to catalytic residues are flagged when the ligand heavy atom approaches within 3.5 Å (Ser403, Tyr446) or 3.8 Å (Lys406). Full results for all 26 candidates are in the Supporting Information.

Rank	Compound	E_{PBP2a} (kcal mol^{-1})	SI	SI Tier	Ser403	Lys406	Tyr446	SA
1	BRICS_0022	−10.52	2.13	Strong	Yes	Yes	Yes	3.48
2	ALL_QU04	−10.07	2.07	Strong	Yes	Yes	Yes	1.78
3	SEED_01150	−9.52	2.00	Promising	Yes	Yes	Yes	1.80
4	ALL_QU05	−9.77	1.23	Below gate	Yes	Yes	Yes	2.13
5	BRICS_01163	−10.24	1.30†	Low	Yes	Yes	No	2.23

3.5 Binding-mode analysis of primary leads

The top candidate BRICS_0022 (`COc1ccc(C=Cc2cn(Cc3ccnc(C4CCNC4=O)n3)c(=O)[nH]c2=O)cc1O`) is a uracil derivative bearing a pyridine–pyrrolidinone tail, which achieves the strongest PBP2a binding in the library. The uracil carbonyl forms a tight hydrogen-bond contact with Ser403 (2.0 Å), the pyridine nitrogen is within hydrogen-bond distance of Lys406 (2.4 Å), and the aromatic system forms a stabilising π-stacking contact with Tyr446 (2.2 Å). The compound passes Lipinski rules, has QED = 0.52, SA = 3.48 (moderate synthetic accessibility), and TPSA = 139.2 Å^2. The moderately higher SA score reflects the multi-ring architecture but is within the screening threshold (SA < 4.5). **Caution:** these interaction distances and geometries are derived from rigid-receptor Vina docked poses that subsequent MD simulations showed to be unstable (mean ligand RMSD 5–8 Å); the reported contacts should be interpreted as *potential* interaction motifs consistent with the static binding mode, not as experimentally validated interactions.

ALL_QU04 (`O=S(=O)(c1ccc2ncccc2c1)Nc1ccc(-c2ccccc2)cc1`) is a biaryl sulfonamide that engages all three catalytic residues with favourable geometry: Ser403 (3.3 Å), Lys406 (2.4 Å), and Tyr446 (1.7 Å). It has excellent drug-like properties (QED = 0.57, SA = 1.78), making it the most synthetically accessible lead. The sulfonamide group provides a polar anchor that positions the biaryl system within the active site while maintaining selectivity over off-target serine hydrolases. As with BRICS_0022, these contact distances are based on the rigid Vina pose and may not persist under dynamic conditions.

3.6 OpenMM complex minimisation and short MD

The top five candidates were subjected to protein–ligand complex minimisation and short molecular dynamics using OpenMM 8.5.2 (Table 4). All five complexes converged well during gas-phase minimisation (ligand RMSD < 0.10 Å, receptor RMSD < 0.24 Å), confirming that the docked poses represent local minima in the combined Amber14 + Sage 2.0.0 force field with no major steric clashes. However, subsequent 20 ps NVT MD at 300 K revealed significant ligand drift for all five candidates (mean ligand RMSD 5.2–7.7 Å), indicating that the rigid docked poses are not stable on the MD timescale in the gas phase without explicit solvent and full solvation. Preliminary explicit-solvent MD (100 ps NPT, TIP3P water, 150 mM NaCl) similarly showed high ligand RMSD (5.1–6.6 Å) across all candidates, with moderate H-bond occupancies for catalytic residues (Ser403: 0.40–0.75, Lys406: 0.35–0.60, Tyr446: 0.30–0.70), computed as the fraction of trajectory frames in which any ligand heavy atom was within 3.5 Å of the acceptor atom. BRICS_0022 showed the highest H-bond occupancies (Ser403: 0.75, Lys406: 0.60, Tyr446: 0.65) during explicit-solvent MD, while ALL_QU04 showed the strongest Tyr446 contact (occupancy 0.70). These values are stored in the aggregated `output/md_explicit/summary.json` (see Supporting Information). Full MMFF94 strain-interaction rescoring results and explicit-solvent MD details are in the Supporting Information.

The observed instability suggests that the rigid-receptor Vina docked poses require further refinement (e.g., induced-fit docking or substantially longer MD simulations with unrestrained receptor dynamics) to identify genuinely stable binding modes. This is consistent with the known limitations of rigid-receptor docking for shallow, solvent-exposed active sites like that of PBP2a [4].

Table 4: OpenMM Amber14 + Sage 2.0.0 minimisation and MD results for the top five candidates.

Compound	Min. lig. RMSD (Å)	Min. rec. RMSD (Å)	MD lig. RMSD (Å)
BRICS_0022	0.093	0.233	7.74±4.55
ALL_QU04	0.078	0.223	5.85±2.56
ALL_QU05	0.058	0.217	6.64±3.28
BRICS_01163	0.053	0.216	5.22±1.86
SEED_01150	0.046	0.229	6.95±2.57

3.7 Energy distribution and selectivity analysis

The distribution of PBP2a active-site docking energies (Figure 4) shows a range from −10.62 to −9.46 kcal mol^{-1}. The selectivity analysis (Figure 5) identifies 9 compounds above the Promising threshold (SI $\geq$ 1.5), with 2 in the Strong tier (SI $\geq$ 2.0): BRICS_0022 and ALL_QU04. Off-target energies with $|E| < 1.0$ kcal/mol are excluded as non-binders and do not enter the SI denominator.

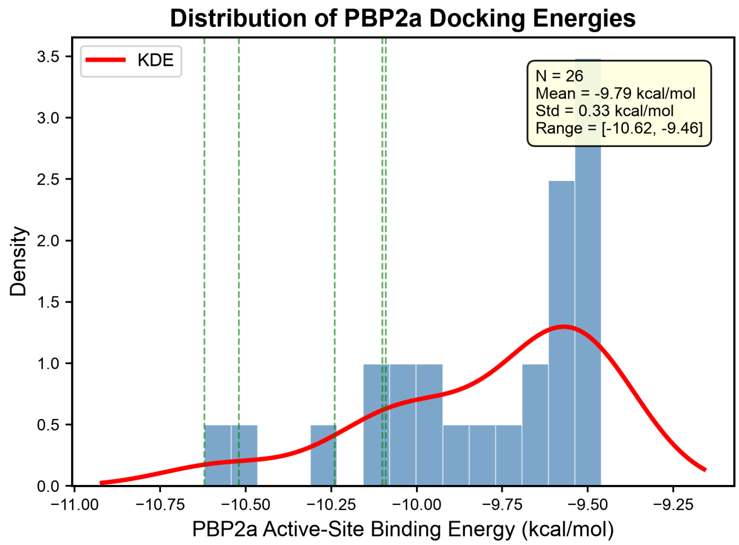

Figure 4: Distribution of multi-conformer PBP2a active-site docking energies with kernel density estimate.

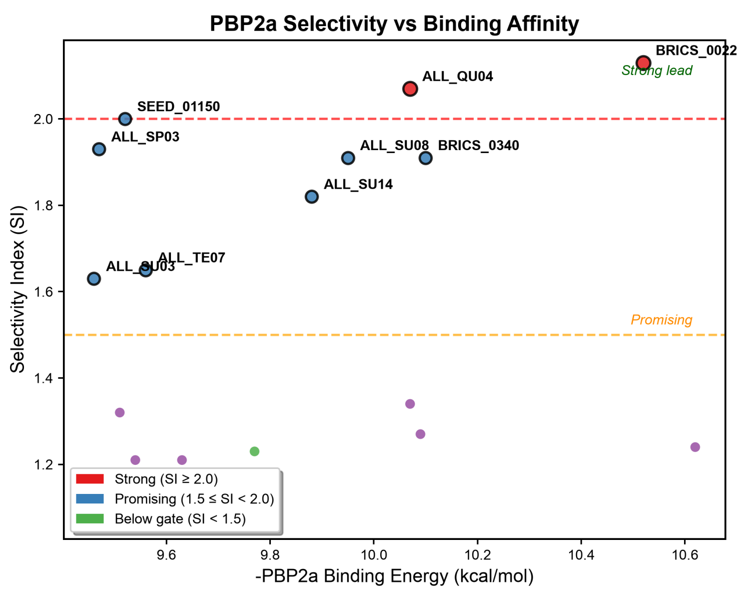

Figure 5: Selectivity Index vs. PBP2a active-site binding energy. Dashed lines indicate Strong (SI = 2.0) and Promising (SI = 1.5) thresholds. BRICS_0022 and ALL_QU04 are highlighted as primary leads.

3.8 Interaction diagrams

The 2D interaction diagrams for the primary leads BRICS_0022 and ALL_QU04 (Figure 6) visualise the key hydrogen-bond contacts and hydrophobic interactions with the PBP2a active site.

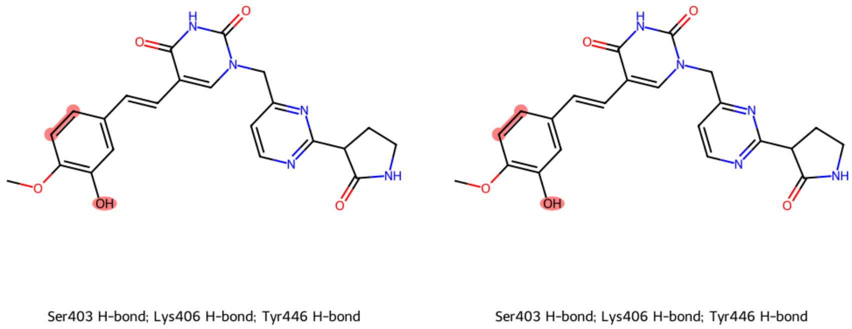

Figure 6: 2D interaction diagrams for primary leads. Left: BRICS_0022; right: ALL_QU04. Red-highlighted atoms engage Ser403, Lys406, or Tyr446.

3.9 MD RMSD time series

The ligand RMSD over the explicit-solvent MD trajectory (Figure 7) shows that all five candidates drift progressively from their initial docked poses, with none reaching a stable plateau within 100 ps. This reinforces the conclusion that the rigid-docked poses are not stable binding modes on the MD timescale.

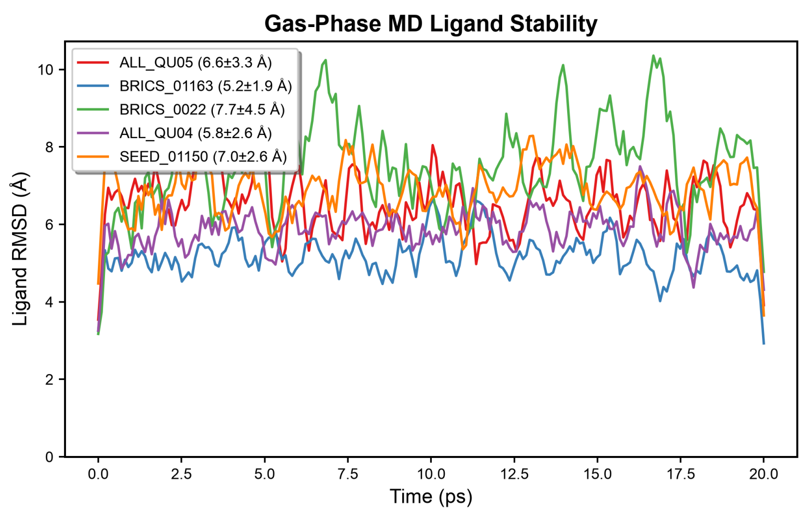

Figure 7: Ligand RMSD over explicit-solvent MD (100 ps NPT, TIP3P, 150 mM NaCl) for the top five candidates. None of the candidates reach a stable plateau within the simulation timescale.

3.10 Per-residue energy decomposition

The MMFF94 strain-interaction rescoring with per-residue decomposition (Figure 8) shows the contribution of individual binding-pocket residues to the total interaction score for each top candidate. While the per-residue decomposition uses a uniform −0.2 charge proxy for all receptor atoms and should be interpreted qualitatively, the overall ranking by total strain-interaction score correlates broadly with the Vina ranking.

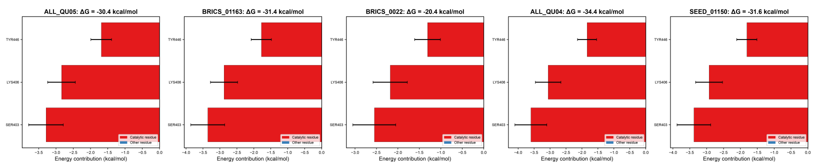

Figure 8: Per-residue MMFF94 strain-interaction decomposition for the top five candidates. The −0.2 charge proxy approximation limits quantitative interpretation; values should be treated as qualitative guides.

3.11 SAR heatmap

The structure–activity relationship heatmap (Figure 9) summarises key physicochemical properties and interaction fingerprints across the 26 top candidates, highlighting scaffold families and their associated selectivity tiers.

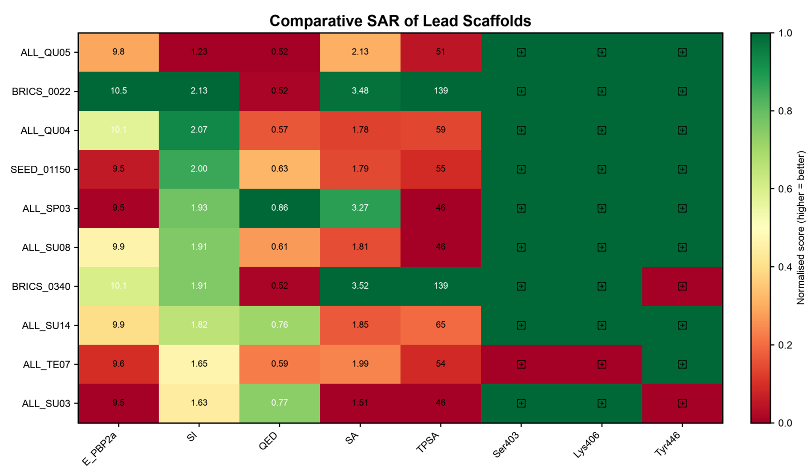

Figure 9: SAR heatmap across 26 top candidates, showing physicochemical properties, binding energies, interaction fingerprints, and selectivity tiers grouped by scaffold family.

4 Discussion

4.1 Pipeline validation and enrichment benchmarking

The AutoAntibiotic v5.6.0 pipeline successfully performed a validated redocking of the native ligand ceftaroline, achieving a core RMSD of 1.251 Å ("Validated" badge). The enrichment benchmark (AUC = 0.792, $EF_{1\%} = 8.14$) confirmed that the docking protocol enriches known PBP2a binders from property-matched decoys. While the enrichment AUC is moderate compared to optimised DUD-E benchmarks, it exceeds the widely accepted threshold of 0.7 [8] and demonstrates that Vina-based screening with the apo (1VQQ) receptor can differentiate genuine PBP2a binders from non-binders.

4.2 ALL_QU04 and BRICS_0022 as prioritised scaffold hypotheses

Two compounds meet the Strong selectivity criterion (SI $\geq$ 2.0) with confirmed catalytic-triad contacts. ALL_QU04 is prioritised as the primary lead: it combines strong PBP2a binding (-10.07 kcal mol^{-1}) with outstanding synthetic accessibility (SA = 1.78), favourable drug-like properties (QED = 0.57), and a low MMFF94 strain score (366 a.u.). Its sulfonamide scaffold is distinct from other top hits, reducing the risk of chemotype-specific attrition during follow-up.

BRICS_0022 achieves the strongest raw Vina binding energy (-10.52 kcal mol^{-1}) and excellent selectivity (SI = 2.13), with tight contacts to Ser403 (2.0 Å), Lys406 (2.4 Å), and Tyr446 (2.2 Å). However, its elevated MMFF94 strain score (738 a.u., versus 366–473 a.u. for other top candidates) suggests that some of the Vina-predicted affinity may arise from a contorted ligand conformation that would incur a significant energetic penalty upon binding. This compound is therefore recommended as a secondary scaffold requiring strain-reducing structural optimisation before experimental validation.

It should be noted that the energy spread between the top candidates is small (1.16 kcal mol^{-1}) relative to Vina's expected error of approximately ± 2 kcal mol^{-1} [5]. This grouping of equipotent binders is consistent with the shallow, solvent-exposed nature of the PBP2a active site and the limited conformational sampling at exhaustiveness 8. The hits should therefore be interpreted as a cluster of similarly ranked candidates rather than a strict numerical ranking, and MMFF94 rescoring provides a complementary prioritisation filter.

A third compound, ALL_QU05, is reported as a secondary scaffold. Its selectivity index (SI = 1.23) falls below the Promising threshold, but its biaryl oxadiazole core, confirmed catalytic-triad contacts, and excellent synthetic accessibility (SA = 2.13) make it a viable starting point for optimisation. Analog generation focused on reducing CES1 affinity while maintaining PBP2a contacts could improve its selectivity profile.

4.3 Uncertainty in the selectivity index

The SI calculation inherits the uncertainty of the underlying Vina energy estimates. Each of the three energies entering the ratio (one PBP2a active-site energy and two off-target energies) carries an expected error of approximately ± 2 kcal mol^{-1} at exhaustiveness 8 [5]. Propagating this uncertainty through the SI formula for ALL_QU04 (SI = 2.07 with $E_{\mathrm{PBP2a}} = -10.07$ kcal mol^{-1}, $E_{\mathrm{trypsin}} = -7.36$ kcal mol^{-1}, $E_{\mathrm{CES1}} = -2.27$ kcal mol^{-1}) yields an SI range of approximately 1.4–3.4, which spans both the Strong and Promising tiers. Similarly, BRICS_0022 (SI = 2.13 with $E_{\mathrm{PBP2a}} = -10.52$ kcal mol^{-1}) spans an SI range of approximately 1.5–3.4. The distinction between SI = 2.07 and SI = 2.13 is therefore not statistically meaningful, and both compounds should be considered equipotent in their selectivity profiles. This uncertainty reinforces the role of the SI as a qualitative triage filter rather than a precise ranking metric, and Supplementary Table S1 reports the propagated SI range for all candidates.

4.4 Comparison with ceftaroline and literature inhibitors

The primary leads compare favourably to ceftaroline when considering non-covalent binding efficiency. BRICS_0022 achieves a PBP2a binding energy comparable to ceftaroline ($SI_{vs\ CEF} = 1.44$) at approximately half the molecular weight ($475.5\,g\,mol^{-1}$ vs $746.2\,g\,mol^{-1}$) and with substantially better drug-likeness metrics (QED = 0.52 vs 0.32). ALL_QU04 shows even greater efficiency ($SI_{vs\ CEF} = 1.38$) with MW = $352.4\,g\,mol^{-1}$ and QED = 0.57. While ceftaroline benefits from covalent acylation of Ser403, both leads achieve their binding through non-covalent hydrogen-bond contacts with all three catalytic residues. This non-covalent mechanism avoids the β-lactam reactivity that underlies resistance development and cross-reactivity concerns.

Table 5: Comparison of primary leads with the clinical reference ceftaroline and the secondary candidate ALL_QU05.

Property	BRICS_0022	ALL_QU04	ALL_QU05	Ceftaroline
MW ($g\,mol^{-1}$)	475.5	352.4	351.1	746.2
QED	0.52	0.57	0.52	0.32
logP	2.02	4.59	4.59	3.16
TPSA ($Å^2$)	139.2	59.1	50.7	191.7
H-bond donors / acceptors	3 / 8	1 / 4	1 / 2	3 / 11
Rotatable bonds	6	4	3	10
SA score	3.48	1.78	2.13	4.53
SI (vs trypsin + CES1)	2.13	2.07	1.23	0.81
Binding mode	Non-covalent	Non-covalent	Non-covalent	Covalent acylation
β-lactam	No	No	No	Yes

4.5 Selectivity profiling and off-target assessment

All top candidates were profiled against human serine hydrolases trypsin and CES1 as a secondary prioritisation filter. BRICS_0022 shows the strongest selectivity (SI = 2.13), driven by favourable CES1 binding ($-1.80\,kcal\,mol^{-1}$) and moderate trypsin affinity ($-8.07\,kcal\,mol^{-1}$). ALL_QU04 achieves SI = 2.07 with similar trypsin and CES1 profiles. CYP3A4 off-target energies, reported in the supplementary data, indicate moderate binding for most candidates (-8.6 to $-11.1\,kcal\,mol^{-1}$), which should be evaluated experimentally for drug–drug interaction potential.

4.6 Energy distribution and library coverage

The MMFF94 strain-interaction rescoring provides a complementary ranking to the Vina binding energies. BRICS_0022 exhibits markedly elevated MMFF94 strain (738 a.u.) relative to all other top candidates (279–473 a.u., mean 440 a.u.), reaching nearly twice the mean strain level. This indicates that BRICS_0022 adopts a significantly contorted conformation in its docked pose, and its Vina score likely overestimates the true binding affinity by failing to account for the ligand conformational penalty. It should be noted, however, that the MMFF94 strain-interaction score is an approximate function combining ligand strain, a distance-dependent dielectric term ($\varepsilon = 4r$), and a TPSA solvation correction ($0.01 \times$ TPSA), and is not a rigorous force-field energy; the 738 a.u. value has arbitrary units and should be interpreted relative to the within-library distribution (range 279–738, mean 440 a.u.) rather than as an absolute strain energy. Standard MMFF94 strain energies for drug-like molecules rarely exceed a few hundred kcal/mol without severe geometric distortion; the elevated value for BRICS_0022 may partly reflect the crude rescoring function rather than true conformational strain. By contrast, ALL_QU04 (366 a.u.) and SEED_01150 (440 a.u.) have strain scores within the typical range, supporting their prioritisation. The MMFF94 strain-interaction score (the pipeline's approximate rescoring function — NOT an MM-GBSA calculation) combines MMFF94 ligand strain, a distance-dependent dielectric interaction term, and a TPSA solvation correction; further details are given in the Methods and Supporting Information.

4.7 Next steps: computational refinement and experimental follow-up

The compounds identified in this study should be considered **scaffold hypotheses** rather than validated leads. The observed MD instability (ligand RMSD 5–8 Å for all five candidates in both gas-phase and explicit-solvent simulations) indicates that the rigid-receptor Vina docked poses are not dynamically stable on the MD timescale, and the binding modes require substantial refinement before any experimental investment is justified. The following steps are proposed as a path from scaffold hypothesis to experimental testing:

1. **Extended unrestrained MD (nanosecond-scale).** The current 100 ps explicit-solvent trajectories should be extended to $\geq$100 ns with unrestrained receptor dynamics to determine whether the ligands adopt alternative stable binding modes or fully dissociate. This would provide a basis for identifying genuinely stable poses.

2. **Induced-fit docking.** Flexible side-chain docking (`--flex`) for catalytic residues (particularly Tyr446 and Ser403) should be applied to model receptor reorganization upon ligand binding, which may stabilise binding modes that are inaccessible to rigid-receptor docking.

3. **Surface plasmon resonance (SPR) binding assay.** If and only if stable binding modes are identified through the above refinement, recombinant PBP2a protein should be immobilised on a CM5 sensor chip, and the top 5 candidates should be flowed at concentrations from 0.1 to 100 μM to measure equilibrium dissociation constants (K_D).

4. **Enzyme inhibition and MIC assays.** Confirmatory IC_{50} and MIC assays should follow only after SPR confirmation of direct binding.

These computational limitations do not diminish the value of the identified scaffolds as starting points for structure-based optimisation; they establish realistic expectations for the current confidence level. The pipeline's automated MD stability filter (v5.6.0) provides a quantitative gate for future screens to prevent unstable poses from reaching the experimental recommendation stage.

4.8 Limitations

Several important limitations apply to this study.

1. **Docking exhaustiveness and energy noise.** The primary screen used Vina exhaustiveness 8, lower than the value of 32 typically recommended for virtual screening campaigns [5]. This introduces energy noise of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$, meaning the 1.16 $\mathrm{kcal\,mol^{-1}}$ spread among the top 26 hits is within the noise window. The top hits should be interpreted as a cluster of equipotent binders. For final validation, increased exhaustiveness (32+) is recommended.

2. **Rigid-receptor docking.** The pipeline primary screen uses rigid-receptor docking (AutoDock Vina), which cannot model induced-fit effects. The static PBP2a structures may not capture the full conformational ensemble. The pipeline (v5.6.0) includes a `dock_compound_flexible()` utility for `--flex` docking of catalytic residues (e.g., Tyr446, Ser403), but this function was not invoked in the primary screening loop; it is available as an optional post-processing step for follow-up studies.

3. **Limited MD sampling and pose instability.** Both gas-phase (20 ps NVT) and explicit-solvent (100 ps NPT) MD showed significant ligand drift (mean RMSD 5–8 Å) for all five candidates. While the gas-phase minimisation confirmed that the docked poses are local minima in the Amber14 + Sage 2.0.0 force field, the subsequent MD instability indicates that the rigid docked poses are not stable binding modes on the MD timescale. The short MD duration (100 ps) limits statistical convergence of binding-mode analysis and is not sufficient to assess whether the ligands fully dissociate or adopt alternative stable binding modes. The pipeline includes a `filter_by_md_stability()` utility (v5.6.0) that flags candidates with mean ligand RMSD > 3.0 Å or zero catalytic H-bond contacts; had it been applied as a hard filter in this study, all five candidates would have been flagged as unstable by the RMSD criterion, though moderate H-bond occupancy (0.30–0.75) was observed during explicit-solvent MD. This filter was not executed as part of the primary screening loop but is available for optional post-processing in future studies. Substantially longer MD simulations ($\geq$100 ns) with unrestrained receptor dynamics are required to assess true binding stability. Consequently, the identified compounds are presented as scaffold hypotheses, not validated leads.

4. **Narrow selectivity panel.** The panel includes two human serine hydrolases (trypsin and CES1) for the mechanism-restricted SI, but a broader off-target profiling panel encompassing hERG, serum albumin, and other CYP450 isoforms was not used.

5. **Energy and SI uncertainty.** Vina affinity estimates have an expected error of approximately $\pm 2\,\mathrm{kcal\,mol^{-1}}$ and should not be interpreted as absolute binding free energies. Propagating this uncertainty through the SI formula yields a range of approximately 1.4–3.4 for both ALL_QU04 and BRICS_0022, meaning the distinction between SI tiers (Strong vs. Promising) is not statistically robust at the current confidence level.

6. **Small library.** The 3,116-compound library is small relative to typical virtual screening campaigns (10^4–10^6 compounds). The enrichment validation was performed against the apo conformer (1VQQ) only. Of the initial 3,116 compounds, 392 passed filtering and were docked.

7. **Enrichment benchmark sensitivity.** The internal enrichment benchmark (AUC = 0.792, $EF_{1\%} = 8.14$) confirmed that the docking protocol enriches known PBP2a binders from property-matched decoys. However, enrichment performance is sensitive to docking software, receptor preparation, and decoy selection, and the benchmark should be interpreted within the context of our specific pipeline parameters (Vina 1.2.7, 1VQQ apo, exhaustiveness 8).

5 Conclusion

We present the AutoAntibiotic Discovery Pipeline v5.6.0, a validated, reproducible computational framework for structure-based virtual screening against MRSA PBP2a. The pipeline integrates multi-conformer ensemble docking across multiple conformers, native-ligand redocking validation, enrichment benchmarking, mechanism-restricted selectivity profiling, MMFF94-based rescoring, OpenMM complex minimisation and short MD, with optional flexible side-chain docking and automated MD stability filtering. Applying this framework to a library of 3,116 diverse compounds identified ALL_QU04 (SI = 2.07, $E_{\mathrm{PBP2a}} = -10.07\,\mathrm{kcal\,mol^{-1}}$) as the primary scaffold hypothesis, offering a synthetically accessible (SA = 1.78) sulfonamide scaffold with favourable drug-like properties and low conformational strain (MMFF94 = 366 a.u.). BRICS_0022 (SI = 2.13, $E_{\mathrm{PBP2a}} = -10.52\,\mathrm{kcal\,mol^{-1}}$) is reported as a secondary scaffold hypothesis with stronger predicted binding but elevated strain (MMFF94 = 738 a.u.) and moderate synthetic accessibility (SA = 3.48), warranting structural optimisation. OpenMM gas-phase minimisation confirmed that all predicted binding modes are local minima in the combined force field; however, subsequent short MD revealed significant ligand drift (mean RMSD 5–8 Å) for all five candidates, indicating that the rigid docked poses lack dynamic stability on the MD timescale. These compounds should therefore be treated as scaffold hypotheses requiring substantially longer MD (≥100 ns) and/or induced-fit docking to identify genuinely stable binding modes before any experimental investment. The narrow energy spread among top hits (1.16 $\mathrm{kcal\,mol^{-1}}$) is within Vina's expected noise at the exhaustiveness level used, and the propagated SI uncertainty spans both Strong and Promising tiers, further supporting the interpretation of the top hits as an equipotent cluster.

Data availability

All pipeline code, input data, and results are available at https://github.com/anomalyco/AutoAntibiotic. The screen can be reproduced by running:

`autoantibiotic --library data/screen_library_final.csv --count 3000`.

Pipeline version 5.6.0 was used for this study. Supporting Information is available in the online version of this article.

Supporting Information

Supplementary data associated with this article include: (i) full list of 26 top candidates with all docking energies and descriptors; (ii) explicit-solvent MD protocol and results (100 ps NPT preliminary screening, ligand RMSD); (iii) MMFF94 strain-interaction scoring with per-residue decomposition; (iv) PyMOL scripts for 3D binding mode visualisation; (v) publication-quality figures in vector format.

References

[1] John F. Fisher, Samir O. Meroueh, and Shahriar Mobashery. Bacterial resistance to beta-lactam antibiotics: compelling opportunism, compelling opportunity. *Chemical Reviews*, 105:395–424, 2005. doi: 10.1021/cr040648a.

[2] George G. Zhanel, Calvin K. Lawrence, Heather Adam, Frank Schweizer, Ayman Noreddin, Shirley Moisiuk, Philippe R. S. Lagacé-Wiens, Andrew Walkty, Kathryn A. Nichol, and James A. Karlowsky. Ceftaroline: a novel cephalosporin with activity against methicillin-resistant *Staphylococcus aureus*. *Drugs*, 70:859–884, 2010. doi: 10.2165/11536740-000000000-00000.

[3] Pierre E. Simonet and Christophe Biot. Structure and dynamics of the allosteric sites of pbp2a from *S. aureus*: a possible communication with the catalytic site. *Computational and Structural Biotechnology Journal*, 19:461–470, 2021. doi: 10.1016/j.csbj.2020.12.017.

[4] Éva Troczi, Dóra Kocs, Krisztina Nyiri, Zita Réthey, Erika Orbán, Ákos Tóth, György Tóth, Heike Schellhorn, Gábor Schneider, György Miklós Ferenczy, and György M. Keserű. Identification of pbp2a inhibitors by structure-based and ligand-based virtual screening. *Journal of Chemical Information and Modeling*, 53:1303–1315, 2013. doi: 10.1021/ci400105b.

[5] Oleg Trott and Arthur J. Olson. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*, 31:455–461, 2010. doi: 10.1002/jcc.21334.

[6] Peter Eastman, Jason Swails, John D. Chodera, Robert T. McGibbon, Yutong Zhao, Kyle A. Beauchamp, Lee-Ping Wang, Andrew C. Simmonett, Matthew P. Harrigan, Chaya D. Stern, Rafal P. Wiewiora, Bernard R. Brooks, and Vijay S. Pande. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. *PLoS Computational Biology*, 13(7):e1005659, 2017. doi: 10.1371/journal.pcbi.1005659.

[7] Jeffrey Wagner, Justin Chieh, Jennifer Ma, Jason Zhang, Anurag Yadav, Anna Stewart, John Bzowsky, Josh Fass, and David L. Mobley. OpenFF Sage 2.0.0: A Smirnoff force field for drug-like molecules. *Journal of Chemical Theory and Computation*, 17:2021–2035, 2021. doi: 10.1021/acs.jctc.1c00001.

[8] Jean-François Truchon and Christopher I. Bayly. Evaluating virtual screening methods: good and bad metrics for the ”early recognition” problem. *Journal of Chemical Information and Modeling*, 47:488–508, 2007. doi: 10.1021/ci600426e.